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Exam question from Amazon's AWS Certified AI Practitioner AIF-C01

Question #: 3

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[All AWS Certified AI Practitioner AIF-C01 Questions]

A company wants to classify human genes into 20 categories based on gene characteristics. The company needs an ML algorithm to document how the inner mechanism of the model affects the output. Which ML algorithm meets these requirements?

A. Decision trees

B. Linear regression

C. Logistic regression

D. Neural networks

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by awsfriend at *Nov. 5, 2024, 12:03 a.m.*

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Comments

Jessiii Highly Voted 1 year, 1 month ago

Selected Answer: A

Decision trees provide a transparent and interpretable model. They allow you to understand how decisions are made by following the path of

splits at each node, which directly shows the influence of input features (gene characteristics) on the final output (gene category). This helps document and explain how the model arrives at its classifications, satisfying the requirement for transparency regarding the inner mechanism of the model.

upvoted 10 times

Salmariaz Most Recent 2 months, 4 weeks ago

Selected Answer: A

Linear regression produces continuous numerical outputs, and logistic regression outputs probabilities.
Because the requirement is to explain the decision logic through clear rules, a (A) decision tree is the most suitable choice, unlike neural networks which lack interpretability.

upvoted 1 times

Clio_Siyi 1 year, 4 months ago

Selected Answer: A

A is correct. I firstly thought Logistic regression should be right, but it's for binary classification, and not suitable for the case in this question because there are 20 categories.

upvoted 3 times

jove 1 year, 4 months ago

Selected Answer: A

Decision Trees can handle multi-class classification problems, making them suitable for categorizing genes into 20 distinct classes

upvoted 2 times

awsfriend 1 year, 5 months ago

Decision trees is correct.

upvoted 3 times